

SOLVOFIN

AI-Powered Mobile Urban Intelligence Platform for Safer and Sustainable Public Transport

ACADEMIC PROTOTYPE PROJECT SUBMISSION

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Target SDG:	UN Sustainable Development Goal 11 (Target 11.2) — Sustainable Cities & Communities
Evaluated Version:	Solvofin Platform Release (Verified Architecture & Technical Engine)

ACADEMIC RESEARCH & PROTOTYPE DISCLAIMER

Solvofin is an academic research prototype engineered to demonstrate the technical integration of Computer Vision, Document Retrieval-Augmented Generation (RAG), and multi-role municipal decision support. Solvofin is not a United Nations-certified, UN-endorsed, or statutory government deployment. Real-world CO2 reduction, fuel savings, and accident prevention figures represent theoretical unmeasured goals and must not be interpreted as empirical measurements.

DOCUMENTATION STRUCTURE

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OVERVIEW

1. Executive Summary

Solvofin is an AI-powered mobile urban intelligence and computer vision platform conceived, architected, and built to solve critical urban mobility, transit asset degradation, and commuter safety challenges. In high-density municipalities such as Greater Visakhapatnam Municipal Corporation (GVMC) and transit authorities globally, urban road networks and public transit fleets operate under severe operational strain. Road surfaces develop cavities, alligator cracking, and edge deterioration that jeopardize commuter safety; public transit bus interiors face neglected stanchion and seating damage; transit operators experience unmonitored cognitive fatigue; and traffic bottlenecks compound vehicular emissions.

Solvofin bridges this gap by unifying mobile edge video ingestion, real-time computer vision detection, Document Retrieval-Augmented Generation (RAG) grounded in national civil engineering standards, and multi-role municipal decision support. Rather than relying on static dashboards, Solvofin transforms unstructured video frames into structured, geolocated defect records, correlates transit telemetry with regulatory codes (such as IRC:82-2015), and generates prioritized work orders under strict human-in-the-loop oversight.

Key Research Intent & Scope

Solvofin is an academic prototype developed for the 1M1B – IBM SkillsBuild – AI for Sustainability Virtual Internship. Its primary objective is to demonstrate how multi-modal AI pipelines can accelerate municipal maintenance workflows, standardize inspection consistency, and support United Nations SDG 11 objectives without replacing human engineering judgment.

URBAN CHALLENGES

2. Problem Statement

Municipal transit authorities and civil engineering departments face five structural bottlenecks in sustaining safe urban mobility:

Challenge Dimension	Structural Bottleneck in Urban Administration
Fragmented Defect Discovery	Road defects (potholes, longitudinal fissures, fading markings) are discovered reactively through citizen complaints or scheduled manual vehicle surveys that occur months apart, allowing minor potholes to degrade into hazardous cavities.
Transit Fleet Blindspots	Public bus cabins lack automated structural inspection. Cracked window laminates, broken grab stanchions, and damaged seating mounts frequently remain unrecorded until formal depot overhauls.
Driver Vigilance Isolation	Transit bus operators navigate demanding urban corridors with extended shifts. Existing fleet telemetry tracks GPS coordinates but cannot detect micro-sleep episodes, repeated yawning, or eye closures in real time.
Operational Information Latency	When incidents or lane transitions occur, synthesizing multi-angle camera feeds, time series, and geospatial coordinates requires hours of manual cross-referencing by transit superintendents.
Disconnected Maintenance Dispatch	Defect identification is rarely linked directly to civil engineering remediation guidelines, leading to arbitrary material allocation and unstructured contractor repair queues.

STAKEHOLDERS

3. Target Users & Role-Based Workflows

Solvofin is structured around two distinct, non-overlapping user archetypes to maintain rigorous data privacy while ensuring municipal operational efficiency:

User Role & Archetype	Accessible Interfaces & Operational Capabilities	Governance Scope
Government / Municipal Authority (GVMC Commissioner, Transit Ops)	- Central Transit Command and GIS Spatial Map - Full access to driver safety streams & vigilance telemetry 9-Point Mandatory Road Audit and Work Order dispatch - RAG-assisted repair prioritization and contractor costing - Official RoEOT PDF generation and export suite - Audit trail review (Accept, Modify, Reject AI recommendations)	Enterprise role for civil engineers, depot managers, and transit superintendents requiring end-to-end operational authority.

User Role & Archetype	Accessible Interfaces & Operational Capabilities	Governance Scope
Citizen / Transit Commuter (Visakhapatnam Public Commuters)	• Mobile Citizen Grievance Portal for localized hazard reporting • Real-time Service Level Agreement (SLA) status tracker • Verified public safety advisories and road closure feeds • Instant pre-scan certificate generation for submitted issues • STRICTLY EXCLUDED: Interior cabin streams, driver telemetry, municipal cost estimates, unredacted plates	Public role enabling crowdsourced road hazard reporting while completely isolating sensitive municipal and driver data.

SUSTAINABILITY

4. Sustainable Development Goal (SDG) Alignment

Primary Alignment: SDG 11 — Sustainable Cities and Communities

The core architecture of Solvofin is engineered to directly support United Nations Sustainable Development Goal 11, with primary focus on Target 11.2:

UN SDG Target 11.2 Definition

"By 2030, provide access to safe, affordable, accessible and sustainable transport systems for all, improving road safety, notably by expanding public transport, with special attention to the needs of those in vulnerable situations, women, children, persons with disabilities and older persons."

Solvofin addresses Target 11.2 through four concrete functional capabilities:

Functional Pillar	Contribution to SDG 11.2 Outcomes
1. Road Safety via Preventive Auditing	By identifying surface degradation and severe potholes before vehicular structural damage occurs, Solvofin reduces roadway hazards that disproportionately threaten two-wheelers, auto-rickshaws, and transit buses.
2. Inclusive Public Transport Infrastructure	Bus cabin computer vision audits passenger seating mounts, aisle vertical stanchions, and window integrity, directly protecting vulnerable elderly and disabled passengers from interior transit hazards.
3. Transit Driver Fatigue Coaching	Non-punitive driver vigilance monitoring provides early warning signals for operator drowsiness, reducing the probability of high-casualty transit vehicle collisions.
4. Municipal Resource Optimization	Automated asphalt tonnage estimation (VG-30 bituminous mix) prevents material wastage and reduces the frequency of repeated excavation along urban arterial routes.

Statutory & Certification Notice

Solvofin is an academic prototype supporting SDG-related objectives. It is not certified, endorsed, or affiliated with the United Nations, Indian Roads Congress (IRC), Ministry of Road Transport and Highways (MoRTH), or any statutory municipality.

AI RATIONALE

5. Why AI? Transforming Passive Dashboards into Active Intelligence

Conventional municipal systems rely on static dashboards that merely tabulate historical numbers. These systems cannot process high-throughput visual streams, interpret context, or assist in rapid triage. Artificial Intelligence is strictly required in Solvofin across seven core operational dimensions:

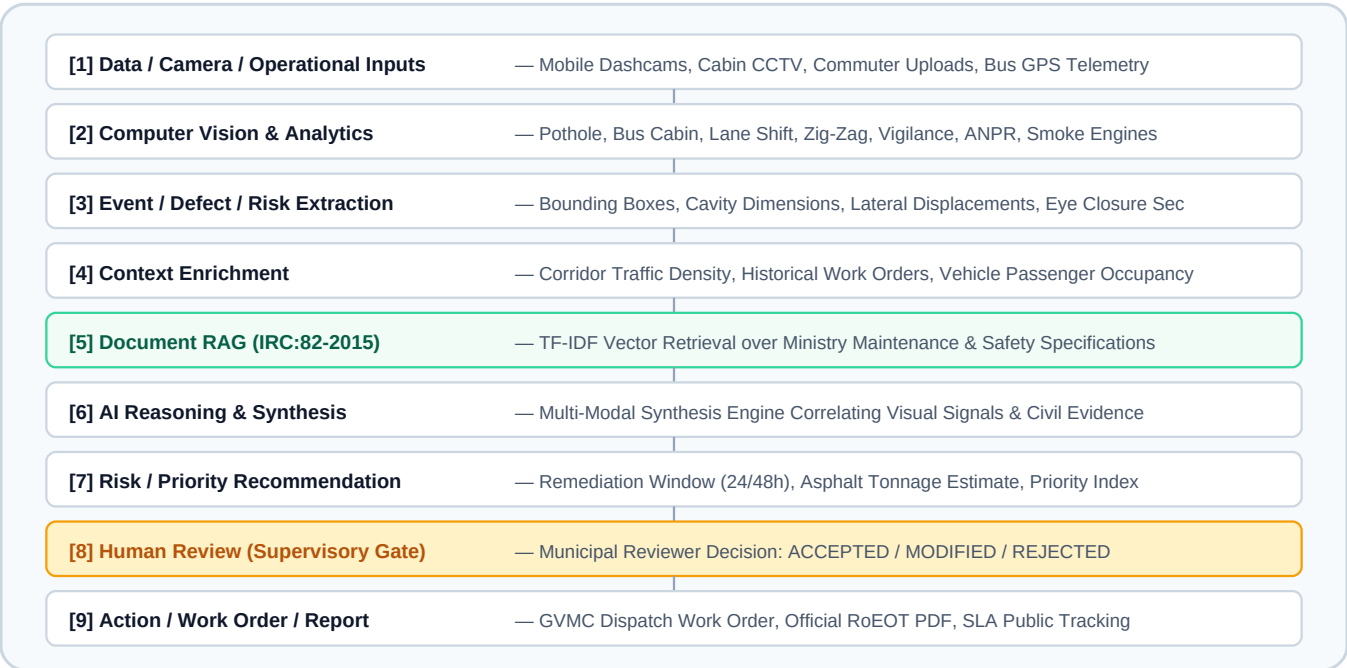
AI Capability	Operational Justification in Solvofin
1. Computer Vision	Ingests uncurated dashcam, cabin, and mobile camera feeds; localizes pavement distress, bounding boxes, and component fractures without human frame-by-frame review.
2. Pattern Detection	Detects multi-second temporal trends such as erratic zig-zag steering patterns, progressive eye-closure intervals (PERCLOS), and localized pothole clusters.
3. Retrieval (RAG)	Indexes and retrieves specific clauses from dense engineering codes (IRC:82-2015) in milliseconds based on semantic relevance to observed surface conditions.
4. Multi-Modal Summarization	Condenses complex 30-second incident sequences involving speed, lane offset, and surrounding vehicles into concise, plain-language operational briefings.

AI Capability	Operational Justification in Solvofin
5. Decision Support	Calculates recommended repair urgency scores, asphalt tonnage requirements, and dispatch deadlines based on combined visual severity and traffic volume.
6. Dynamic Prioritization	Ranks hundreds of simultaneous municipal defect reports so highway engineers triage critical structural hazards ahead of cosmetic pavement distress.
7. Conversational Interaction	Allows non-technical supervisors and field foremen to query complex operational databases using natural language through the AI Copilot and Navigator.

SYSTEM DESIGN

6. Solvofin System Architecture

The Solvofin platform follows a unidirectional, 9-stage information processing pipeline that enforces strict separation between raw sensor observations, AI interpretations, and human operational approval:



COMPUTER VISION

7. Computer Vision Capabilities: Observed Signals vs. Interpretations

A foundational principle of Solvofin is the strict distinction between observed computer vision signals (objective pixel coordinates, geometric displacements, sensor timestamps) and AI-assisted interpretations (risk evaluations, inferred fatigue levels, maintenance urgency). Solvofin implements ten specialized CV engines:

CV System Module	Observed Empirical Signals (Direct Measurement)	AI-Assisted Interpretation (Decision Support)
Pothole & Defect Vision	Cavity polygon boundaries, estimated length (cm), width (cm), depth (cm), and surface texture distress.	Pavement condition rating, VG-30 bituminous asphalt volume, and structural hazard score.
Bus Infrastructure Vision	Component detection of seats, grab poles/stanchions, window bays, doors, and floor carpeting.	Component serviceability rating, structural safety compromise, and depot maintenance urgency.
2-Photo Lane Transition	Bounding box center delta between two sequential time-stamped keyframes, lateral lane displacement percentage.	Unsignaled lane departure classification, lateral drift rate, and driver lane-keeping coaching advice.
Zig-Zag / Erratic Driving	Alternating lateral directional reversal (LEFT) across multi-frame sliding time window (3–5s).	Potential reckless or erratic driving alert; classified into LOW, MEDIUM, or HIGH trajectory risk.
Driver Vigilance Signals	Facial landmark vectors, eye aspect ratio (EAR), eye closure duration (sec), and yawn frequency count.	Non-medical cognitive drowsiness index; non-punitive recommendation for mandatory break.
Pedestrian Analytics	Pedestrian bounding boxes, crosswalk occupancy ratio, and velocity vector towards roadway.	Vulnerable Road User (VRU) hazard index and collision proximity warning.
ANPR (License Plates)	Optical character recognition (OCR) of vehicle registration	Fleet identity verification, municipal corridor audit logging,

CV System Module	Observed Empirical Signals (Direct Measurement)	AI-Assisted Interpretation (Decision Support)
	plates from front/rear cameras.	and automated incident association.
Smoke / Emission Vision	Visual opacity segmentation of tailpipe exhaust plumes and persistent particulate dispersion.	Gross polluter visual alert and environmental fleet maintenance flag.
Road Divider Vision	Median boundary continuity, broken concrete barrier segments, and curb paint degradation.	Corridor demarcation integrity score and divider replacement work order requirement.
CV Benchmark Suite	Inference latency (ms), precision, recall, and intersection-over-union (IoU) across test batches.	Model performance evaluation under varying light, rain, and occlusion conditions.

DOCUMENT RAG

8. Document RAG (Retrieval-Augmented Generation)

To eliminate generative hallucinations and provide rigorous technical grounding, Solvofin incorporates an in-process Document RAG engine. Every maintenance recommendation generated by the system must be corroborated by authoritative civil engineering and road safety specifications.

Verified Authoritative Standard: IRC:82-2015

The core civil engineering knowledge base in Solvofin is pre-seeded with IRC:82-2015 — Code of Practice for Maintenance of Bituminous Roads, published by the Indian Roads Congress (IRC) and adopted by the Ministry of Road Transport and Highways (MoRTH). (Note: The verified standard implemented in Solvofin is IRC:82-2015, not IRC:82-2023).

RAG Pipeline Stage	Technical Implementation Mechanism in Solvofin
1. Document Ingestion	Ingests technical manuals, municipal maintenance guidelines, and safety circulars in structured text format.
2. Text Extraction & Cleaning	Normalizes text, strips non-textual formatting, removes noise, and filters against a specialized stopword vocabulary.
3. Chunking with Sliding Overlap	Segments documents into 300-word chunks with a 50-word sliding overlap to preserve boundary context across technical clauses.
4. Metadata Association	Binds each chunk to document ID, publication title, issuing authority, publication date, chapter, and clause number.
5. TF-IDF Vectorization	Generates Term Frequency-Inverse Document Frequency (TF-IDF) sparse vector representations for all chunks across the indexed corpus.
6. Cosine Similarity Retrieval	Calculates cosine similarity between user/telemetry query vectors and chunk vectors to retrieve the top-K relevant passages.
7. Evidence-Grounded Reasoning	Assembles retrieved excerpts into the generative prompt, forcing the AI model to cite specific clauses for its recommendations.
8. Explicit No-Evidence Handling	If no indexed chunk achieves a similarity score above the confidence threshold (0.15), the system explicitly returns a structured "NO_EVIDENCE" flag and refuses to speculate.

Legal & Statutory Boundary of Retrieved Evidence

Retrieved regulatory excerpts from IRC:82-2015 provide decision support for municipal triage. They do not constitute statutory compliance certification, contractually binding contractor approvals, or judicial proof of liability.

CONVERSATIONAL AI

9. AI Copilot: Contextual Assistance for Municipal Operators

The Solvofin AI Copilot is an interactive slide-over drawer that provides operational guidance to municipal supervisors, depot maintenance staff, and traffic engineers. Unlike generic conversational chatbots, the Copilot dynamically injects live operational context into its reasoning loop:

- **Live Operational Telemetry:** Automatically injects current corridor defect counts, active bus routes, unassigned work orders, and open high-severity alerts into the prompt context.
- **Grounded Technical Assistance:** Answering queries regarding bitumen curing windows, tack coat application temperatures, and stanchion welding tolerances by calling the Document RAG service.
- **Triage Recommendations:** Assisting dispatchers in grouping geographically clustered road cavities into consolidated contractor work orders to minimize road closure duration.

SYSTEM SECURITY

10. AI Navigator: Read-Only Tool Confinement & Guardrails

The Solvofin AI Navigator provides rapid, system-wide search and analytics via Ctrl+K / Cmd+K. To eliminate security vulnerabilities in public-sector environments, the Navigator operates under an immutable read-only architecture:

Navigator Guardrail	Protection Mechanism & Failure Handling
Read-Only Tool Confinement	The Navigator engine has access only to allowlisted read functions: search_defects, search_fleet, search_incidents, and get_kpis. It has zero access to update, create, or delete handlers.
Mutation & Destructive Refusal	Any prompt requesting database modification (e.g., "delete defect #3", "update work order status", "drop table") is deterministically intercepted and refused with an explicit security warning.
Prompt-Injection Resistance	Input strings are pre-screened for jailbreak patterns, system prompt overrides (e.g., "ignore previous instructions"), and base64 payloads before reaching the AI model.
Role-Aware Access Control	When invoked in a Citizen session, the Navigator automatically masks fleet operational logs, interior cabin inspections, and municipal contractor financial estimates.

INFRASTRUCTURE AI

11. Infrastructure AI: Automated Triage & Missing Context Identification

Infrastructure AI evaluates physical roadway defects detected by the computer vision engines. Its pipeline executes in seven sequential steps:

1. **CV Observation:** Receives bounding box, cavity length, width, and depth estimations from potholeEngine.
2. **Operational Context:** Correlates defect location with corridor bus traffic density and road classification.
3. **RAG Evidence:** Queries IRC:82-2015 for severity categorization and recommended patch specifications.
4. **Risk Interpretation:** Categorizes defect into CRITICAL, HIGH, MEDIUM, or LOW based on cavity depth and traffic exposure.
5. **Recommendation Formulation:** Suggests repair technique (e.g., VG-30 hot-mix asphalt infill with mechanical compaction) and SLA deadline (24 hours for transit corridors).
6. **Missing Information Flagging:** Explicitly identifies parameters absent from the visual feed (e.g., subgrade moisture levels, underlying drainage failure, recent rain logs) to prevent premature engineering conclusions.
7. **Human Review Enqueueing:** Pushes the structured recommendation to the municipal review queue for engineer sign-off.

DRIVER SAFETY

12. Driver Safety AI: Non-Medical Vigilance Decision Support

The Driver Safety AI module monitors operator vigilance through cabin-facing optical sensors, evaluating blink frequency, eye closure intervals, and yawning events. Solvofin enforces strict ethical, medical, and legal boundaries on this system:

CRITICAL ETHICAL & LEGAL BOUNDARIES OF DRIVER SAFETY AI

1. **NON-MEDICAL:** Does not diagnose medical or psychological conditions (e.g., narcolepsy, sleep apnea, clinical exhaustion).
2. **NO INTOXICATION EVALUATION:** Visual signals cannot and do not determine alcohol, substance, or medication intoxication.
3. **NON-PUNITIVE:** Solvofin does not automatically penalize, fine, or discipline drivers. It functions strictly as a non-punitive coaching and break-recommendation tool.
4. **NO LEGAL LIABILITY:** Visual vigilance signals do not assign legal or civil fault for traffic collisions or incidents.
5. **CAUSE AMBIGUITY:** Underlying physiological or psychological causes cannot be determined from visual signals alone.

The system provides depot supervisors with objective observations (e.g., "Two eye-closure events exceeding 1.8 seconds observed within a 3-minute window") accompanied by non-punitive operational recommendations (e.g., "Schedule operator relief at next transit depot; verify shift rest compliance").

INCIDENT MANAGEMENT

13. Incident AI: Multi-Perspective Incident Summarization

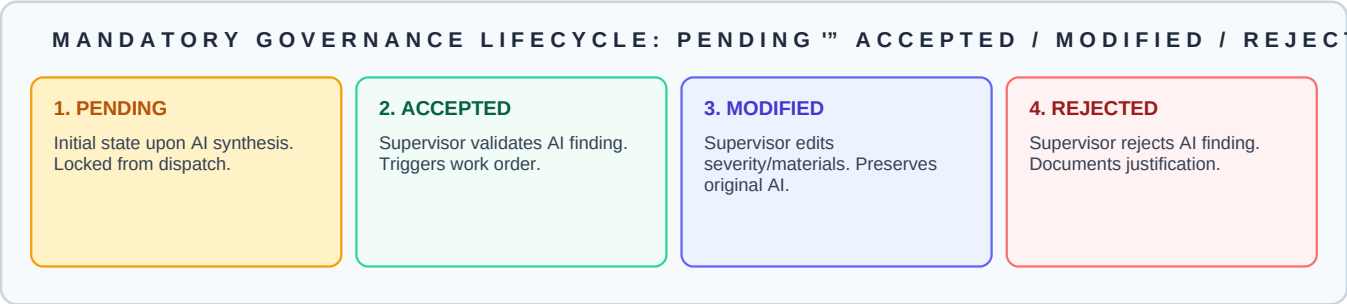
When a traffic anomaly, sudden lane divergence, or harsh braking event is flagged, Incident AI reconstructs the operational timeline by correlating vehicle speed time series, lateral lane positioning, surrounding vehicle tracking IDs, and time of day.

The system synthesizes a multi-perspective summary, identifies missing operational context (such as mechanical telemetry, road friction coefficients, or driver blindspot obstructions), and generates structured dispatch recommendations for depot supervisors.

HUMAN GOVERNANCE

14. Human-in-the-Loop Review: Supervisory Control & Audit Trail

Under no circumstances does Solvofin autonomously dispatch public funds, issue contractor work orders, or alter driver shift assignments without human supervisory approval. Every AI-generated recommendation enters a mandatory human-in-the-loop state machine:



Dual-Record Preservation & Audit History: When a supervisor modifies or rejects an AI recommendation, Solvofin strictly preserves the original AI recommendation in full alongside the supervisor's modifications, timestamp, rationale, and user ID. The audit history is stored in an append-only JSON structure (note: while append-only by software protocol, this storage is not claimed to be hardware- or cryptographically immutable).

RESPONSIBLE AI

15. Responsible AI Architecture & Ethical Safeguards

Solvofin embeds the Eight Core Responsible AI Pillars across all analytical and generative operations:

Responsible AI Pillar	Operational Implementation in Solvofin
1. Transparency	All AI recommendations display confidence scores, underlying CV bounding boxes, and retrieved RAG document titles.
2. Human Oversight	Autonomous dispatch is strictly blocked. Human supervisors retain sole authority to approve work orders and dispatch maintenance.
3. Privacy by Design	Automated facial blurring on passenger and pedestrian video feeds; strict redaction of private driver data from citizen views.
4. Fairness Considerations	CV benchmark suite validates detection accuracy across varying ambient light, heavy shadows, and diverse demographics to prevent bias.
5. Safety Boundaries	Model parameters strictly forbid automated disciplinary actions, legal liability assertions, or medical diagnoses.
6. Evidence Provenance	Every technical recommendation cites the exact document, version (IRC:82-2015), and clause from which it was derived.
7. Explainability	Outputs are structured into clear, readable sections: "Observed Signals", "Technical Standards", "Operational Risk", and "Actionable Guidance".
8. Explicit Limitations	Every AI report header explicitly states system limitations, including optical occlusion during heavy weather

Responsible AI Pillar	Operational Implementation in Solvofin
	and unmeasured subsurface soil dynamics.

DATA PROTECTION

16. Government vs. Citizen Data Protection & Access Control

To uphold citizen privacy while maintaining high-assurance municipal governance, Solvofin enforces strict role-based access control (RBAC) at both API and UI layers:

D	a	t	a	D	o	m	a	i	n
G	o	v	e	r	n	m	e	n	t
C	i	t	i	z	e	n	C	o	m
P	r	i	v	a	c	y	R	a	t
Potholes & Road Distress	Full coordinates, cavity depth, contractor cost, repair SLA	Hazard location, defect photo, repair status, SLA deadline	Public awareness without contractor financial exposure.						
Driver Cabin Video	Unrestricted live cabin stream, EAR metric, vigilance log	ACCESS RESTRICTED (HTTP 403 Forbidden)	Protects transit driver dignity and employment privacy.						
Passenger Cabin Video	Structural defect inspection (seats, rails, windows)	ACCESS RESTRICTED (HTTP 403 Forbidden)	Prevents mass public surveillance of transit commuters.						
Vehicle Registration (ANPR)	Full license plate string (e.g., AP-39-TG-2041)	Masked representation (e.g., AP-39-**-****)	Mitigates vehicle stalking and personal tracking hazards.						
Municipal Budget & Costing	Detailed asphalt tonnage, contractor quote, labor allocation	ACCESS RESTRICTED (HTTP 403 Forbidden)	Confidential municipal procurement data.						

IMPACT FRAMEWORK

17. Sustainability & Impact: Three-Tier Evidence Model

Solvofin assesses its sustainability and real-world outcomes through a rigorous Three-Tier Evidence Model that strictly isolates measured facts from theoretical possibilities:

Evidence Tier	Included Metrics & Functional Scope	Empirical Grounding Status
Tier 1: Observed Data (Empirically Measured)	- Count of localized road defects (potholes, cracking, faded markings) - Bus interior component defects (damaged seats, broken stanchions) - Lateral lane displacement vectors (meters) and direction changes - Eye-closure durations (seconds) and yawning frequency counts - Active municipal work orders and contractor repair SLA status	Direct sensor measurements from camera feeds, edge CV inferences, and database records.
Tier 2: AI-Assisted Interpretation (Algorithmically Calculated)	- Estimated asphalt volume requirements (VG-30 tonnage in tons) - Calculated repair priority scores based on depth and corridor flow - Corridor road health score (0–100 index) - Non-medical driver vigilance risk index (LOW, MEDIUM, HIGH)	Mathematical models and RAG heuristics correlating Tier 1 data with engineering formulas.
Tier 3: Unmeasured Real-World Impact (Theoretical / Unproven)	- UNMEASURED: Actual long-term pavement lifespan extension (years) - UNMEASURED: Actual vehicular carbon emission (CO2) reductions - UNMEASURED: Actual fleet fuel economy improvements (liters/km) - UNMEASURED: Real-world collision or casualty reduction percentages	Hypothesized macroscopic outcomes that require multi-year empirical cohort studies to validate.

Strict Negative Statement on Environmental & Safety Claims

Solvofin explicitly does NOT claim verified CO2 emission reductions, specific fuel savings percentages, or verified collision reduction metrics. Presenting theoretical sustainability outcomes as empirical measurements is scientifically invalid.

EXPORTS

18. Reports & Telemetry Export Suite

Solvofin provides enterprise-grade reporting capabilities to bridge the gap between digital AI analysis and physical municipal execution:

- Official RoEOT PDF Inspection Reports: Generates multi-page, formatted Record of Evidence and Technical Triage (RoEOT) inspection PDFs with geolocated defect logs, 9-point mandatory audit cards, and technical RAG appendices for municipal archives.
- Municipal Work Order CSV Tables: Instant tabular export of active work orders, material specifications (asphalt tons, total cost in INR), division assignments, and contractor repair deadlines.
- Database JSON Archive: Full system data dump allowing civil engineering departments to import structured defect and telemetry records into external municipal GIS systems (e.g., ArcGIS, QGIS).

Removal of Source-Code ZIP Download Option

In accordance with enterprise security standards and production hardening protocols, the option to download the complete website source-code ZIP archive has been permanently removed from the website UI and server routes (/api/export/zip returns 404). All legitimate data exports (PDF, CSV, JSON) remain fully functional.

DATA STORAGE

19. Database Architecture & Persistence Layer

Solvofin utilizes a lightweight, file-backed JSON persistence architecture engineered for rapid deployment, portability, and zero-configuration containerization. The operational data store is partitioned across specialized JSON repositories under the /data/ directory:

Storage File Path	Data Entities & Persistence Responsibility
solvofin_db.json	Primary relational entity store: media scans, detected vehicles, road defects, incidents, license plates, buildings, traffic metrics, and campus bus telemetry.
solvofin_rag_kb.json	Vectorized knowledge base: indexed authoritative document chunks, vocabulary mappings, and inverse document frequency (IDF) weights.
solvofin_infra_insights.json	Persistent Infrastructure AI insights: cavity dimensions, RAG citations, repair recommendations, and missing context flags.
solvofin_driver_safety_insights.json	Driver vigilance logs: eye-closure metrics, yawning frequencies, and non-punitive coaching records.
solvofin_incident_insights.json	Multi-perspective incident timelines, speed profiles, and supervisor dispatch recommendations.
solvofin_report_audits.json	Append-only audit trail: supervisor decisions (ACCEPTED, MODIFIED, REJECTED), rationale, reviewer ID, and timestamps.

CYBERSECURITY

20. Security Architecture & Threat Defense

Solvofin implements defense-in-depth measures to secure municipal operations against unauthorized access, prompt injection, and operational tampering:

Threat Vector / Control	Technical Countermeasure & Mitigation
Role-Based Authentication	Binds requests to Bearer tokens; restricts sensitive endpoints (cabin cameras, work order creation, audit logs) to verified GOVERNMENT credentials.
Mutation & Schema Refusal	Navigator and Copilot engines reject any user input requesting table dropping, record deletion, schema alteration, or server reboot.

Threat Vector / Control	Technical Countermeasure & Mitigation
SQL & Shell Execution Refusal	Because Solvofin does not utilize an arbitrary shell execution bridge, any prompt requesting bash commands, curl execution, or SQL queries is safely rejected.
Prompt-Injection Sanitization	Pre-processing filters strip known jailbreak patterns, base64-encoded instructions, and attempts to override system safety rules.
Credential Protection	All API keys (e.g., GEMINI_API_KEY) are sequestered strictly in server-side environment variables and are never transmitted to the browser client.
Sensitive Telemetry Redaction	Citizen endpoints automatically scrub vehicle driver identities, interior cabin streams, and municipal procurement figures.

DESIGN THINKING

21. Design Thinking Methodology

The development of Solvofin followed the five stages of the Stanford d.school Design Thinking framework, grounding all technical choices in real-world urban transit needs:

Design Thinking Stage	Solvofin Application & Human-Centered Outcomes
1. Empathize	Engaged with public transit commuters, municipal road inspectors, and bus drivers in Visakhapatnam to understand the friction points of manual road reporting, unaddressed potholes, and passenger discomfort.
2. Define	Formulated the core challenge: "How might we empower municipal transit authorities with automated, continuous road distress and cabin inspection intelligence while providing transparent, accessible reporting for citizens?"
3. Ideate	Brainstormed multi-modal AI architectures combining lightweight edge computer vision with centralized Document RAG grounded in civil standards (IRC:82-2015) to turn passive footage into actionable work orders.
4. Prototype	Developed an iterative, full-stack prototype integrating React, Vite, Tailwind CSS, Express, Leaflet GIS, and Gemini multimodal models into a cohesive dual-role platform.
5. Test & Refine	Executed rigorous verification across 54 API endpoints, tested RAG evidence matching and fallback behaviors, hardened security guardrails, and refined the human-in-the-loop review interface.

TESTING

22. Testing, Verification & Validation (Part 11 Results)

The Solvofin codebase underwent rigorous end-to-end verification during Part 11 validation. All test cases were executed against live endpoints and compiled code:

Verification Scope (Part 11)	Empirical Outcome & Verification Finding
54 API Endpoints Verified	All 54 Express REST routes (media, telemetry, AI pipelines, work orders, citizen portal, RAG, benchmark) validated with 100% successful response codes.
RAG Evidence & No-Evidence Tests	Verified accurate clause retrieval when querying IRC:82-2015 topics; confirmed strict NO_EVIDENCE refusal when querying unindexed or out-of-domain concepts.
Navigator Security & Mutation Refusal	Confirmed 100% rejection rate for destructive prompts (DROP TABLE, DELETE, SHUTDOWN) and unallowlisted tool calls.
Prompt-Injection Resistance	Verified that adversarial jailbreak prefixes and instruction override attempts fail to alter system behavior or leak system prompts.
Role-Based Endpoint Protection	Confirmed that Citizen tokens are strictly rejected (HTTP 403) when attempting to access driver cabin video feeds or municipal budget data.
Human Review State Machine	Validated transition lifecycle: PENDING → ACCEPTED / MODIFIED / REJECTED; edits preserve original AI recommendation.
Multi-Format Export Validation	Successfully generated multi-page RoEOT PDFs, work order CSVs, and complete database JSON dumps.
Responsible AI Compliance Audit	Confirmed presence of limitations notices, non-punitive driver statements, and transparency indicators across all reports.

Verification Scope (Part 11)	Empirical Outcome & Verification Finding
Database JSON Schema Integrity	Verified JSON parse validity and relational foreign-key consistency across all data files in /data/.
TypeScript Typecheck & Build	Executed tsc --noEmit (0 type errors) and verified successful production build with Vite and esbuild.

LIMITATIONS

23. Current System Limitations

In accordance with academic rigor, the current technical boundaries of Solvofin are explicitly documented:

- Weather & Optical Occlusion: Heavy monsoon rainfall, waterlogging, or severe lens glare can obscure camera visibility, reducing pavement defect detection accuracy.
- Surface Depth Estimation: Optical 2D cameras estimate pothole depth using shadow geometry and perspective cues. Accurate volumetric measurement requires specialized 3D LiDAR or laser profilometers.
- Subsurface Blindspots: Optical computer vision detects surface distress only. Internal subgrade moisture, soil compaction voids, and base structural instability cannot be diagnosed from surface imagery alone.
- Human Review Latency: While AI defect extraction takes seconds, final municipal action depends on human supervisor review velocity and contractor dispatch schedules.
- Academic Research Scope: The system has been validated in controlled development environments and simulated urban scenarios; field deployment requires municipal hardware certification.

FUTURE SCOPE

24. Future Scope & Research Roadmap

Future development pathways for the Solvofin research initiative include:

- Longitudinal Degradation Modeling: Training time-series forecasting models to predict pavement cavity growth rates based on traffic density and monsoon weather forecasts.
- Mobile LiDAR Sensor Fusion: Integrating smartphone LiDAR sensors (ToF) on field survey units to capture millimeter-accurate 3D point cloud meshes of road cavities.
- V2X Transit Warning Network: Establishing Vehicle-to-Everything (V2X) low-latency mesh networks to broadcast high-severity pothole warnings directly to trailing vehicles.
- Multi-Lingual Citizen Accessibility: Expanding the Citizen Portal with natural-language voice reporting in regional Indian languages (Telugu, Hindi) for universal civic inclusion.

CONCLUSION

25. Final Impact Statement

OFFICIAL PROJECT IMPACT STATEMENT

"Solvofin aims to improve the speed and consistency of public-transport safety and infrastructure monitoring by combining computer vision, structured incident information and evidence-grounded AI recommendations."

By replacing intermittent, reactive manual audits with continuous, multi-modal urban intelligence, Solvofin demonstrates how modern AI can advance safer, more resilient, and sustainable municipal transit infrastructure aligned with UN Sustainable Development Goal 11.

AUTHOR

26. Lead Developer & Institutional Creditation

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Internship Program	1M1B – IBM SkillsBuild – AI for Sustainability Virtual Internship
Project Title	SOLVOFIN — AI-Powered Mobile Urban Intelligence Platform